Symbiosis Institute of Technology, Nagpur

Bachelor of Technology

in

Computer Science and Engineering

"Predictive Analytics and Explainable Machine Learning Framework for Telecom Customer Churn Prediction"

Submitted by

Arya Jain

PRN: 23070521218

Semester: VII

Section: C

Course : Predictive Analytics

Faculty

Prof. Deepak S Asudani

Academic Year: 2026

॥वसुधैव कुटुम्बकम्॥

Symbiosis International (Deemed University), Pune

SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)
(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC
Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

DECLARATION

I hereby declare that the project report submitted herein, titled "Predictive Analytics and Explainable Machine Learning Framework for Telecom Customer Churn Prediction", has been carried out by me as part of the Predictive Analytics course towards the partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering. The work is original and has not been submitted earlier, in whole or in part, for the award of any degree or diploma at this or any other Institution / University.

I hereby assign all copyright rights in this work, including any revised or expanded derivative works, to Symbiosis Institute of Technology, Nagpur. Any content sourced from references, manuals, datasets, or other external materials has been duly acknowledged and appropriately cited.

Name of Student:

Arya Jain (PRN: 23070521218)

Place: Nagpur

Date: October 9, 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Symbiosis Institute of Technology, Nagpur, for providing me with the opportunity to undertake this project as part of the Predictive Analytics course. Working on demand forecasting for supply chain optimization has been a valuable learning experience, helping me develop practical skills in data analysis, predictive modelling, and the interpretation of results.

I am deeply thankful to my course faculty, Prof. Deepak S Asudani, for his valuable guidance, constructive feedback, and encouragement throughout the completion of this project. His teaching and support played an important role in improving my understanding of Knowledge Discovery in Databases, time series forecasting, and the practical application of predictive analytics.

I am immensely grateful to our academic institution, Symbiosis Institute of Technology, Nagpur, for providing the necessary academic environment, resources, and support required to successfully complete this project.

I would also like to express my sincere appreciation to the faculty members of the Department of Computer Science and Engineering. Their guidance and academic support helped me strengthen my understanding of the concepts involved in this project and apply them effectively.

Lastly, I would like to acknowledge my family and friends for their unwavering support, encouragement, and motivation throughout the completion of this project. Their constant encouragement and positive support gave me the confidence to successfully complete the project and overcome challenges along the way.

I sincerely thank everyone who contributed directly or indirectly to the successful completion of this project. Their valuable support, guidance, and encouragement have made this learning experience meaningful and rewarding.

ABSTRACT

Customer Churn prediction - an issue that puts a real dent in the bank for telecommunications providers . When customers just up and cancel their service, it can send revenue and customer retention plans into a tailspin. This project takes on the challenge of predictive analytics using 3 telecom churn datasets : the IBM Telco Customer Churn dataset, a bigger Telecom Data set containing a whopping 51,047 rows and 58 columns - and the Customer Churn Prediction 2020 dataset.

Three classification algorithms got put through their paces separately on each dataset, resulting in nine model experiments. We then compared these experiments using accuracy, precision, recall, F1-score, and ROC-AUC. The best result came from the Random Forest model on the Customer Churn Prediction 2020 dataset, with a very respectable 94.82% accuracy and precision of 88.00%, a less impressive (but still respectable) recall of 73.33%, a steady F1-score of 80.00% and a solid ROC-AUC of 0.8887 on a test split of 850 records.

The saved model got reloaded and gave us the same old metrics all over again, which was reassuring. Plus we got a look at a feature-importance analysis, which showed us that total day charges, customer service calls and total day minutes are the top features. And in a separate test we had the model make 750 new predictions for some unlabeled test records - but we can't compare how well they did because we dont actually know what their labels are. This report tells the story of the workflow, what we found out, what we cant do yet and where we think we can improve.

Keywords: telecom churn, predictive analytics, classification, Random Forest, Logistic Regression, Decision Tree, feature importance, customer retention

CONTENTS

1 Introduction 1
1.1 Background 1
1.2 Problem Statement 1
1.3 Motivation 1
1.4 Objectives 1
1.5 Scope 2
1.6 Significance 2
1.7 Overview of the Report 2
2 Literature Review 3
2.1 Knowledge Discovery in Databases and Predictive Analytics 3
2.2 Customer Churn Prediction 3
2.3 Logistic Regression 3
2.4 Decision Tree and Random Forest 4
2.5 Explainable Machine Learning 4
2.6 Summary and Relevance 4
3 Workflow / Predictive Modelling Process 5
3.1 Overview of the Workflow 5
3.2 Stage 1: Problem Identification and Understanding 5
3.3 Stage 2: Data Selection and Understanding 6
3.4 Stage 3: Data Preprocessing 7
3.5 Stage 4: Feature Preparation 8
3.6 Stage 5: Exploratory Data Analysis 8
3.7 Stage 6: Predictive Modelling 10
3.8 Stage 7: Model Evaluation 10
3.9 Stage 8: Interpretation and Prediction 10
4 Results and Discussion 11
4.1 Model Performance 11
4.2 Final Model Confusion Matrix 11
4.3 Feature Importance 12
4.4 Prediction Demonstration 13
4.5 Practical Implications 13
4.6 Limitations 13
5 Conclusion 15
5.1 Summary of the Work 15
5.2 Key Findings 15
5.3 Future Work 15
6 References 16

LIST OF FIGURES

Figure 1.1: End-to-end workflow of the churn prediction project
Figure 3.1: Record counts for the three analysed datasets
Figure 3.2: Dataset 3 labelled training class distribution
Figure 5.1: F1-score comparison across nine experiments
Figure 5.2: Final Dataset 3 Random Forest metrics
Figure 5.3: Confusion matrix of the final Random Forest
Figure 6.1: Qualitative overview of leading features
Figure 6.2: Predictions for 750 unlabelled records

LIST OF TABLES

Table 2.1: Comparison of the three machine-learning algorithms

Table 2.2: Evaluation metrics and their interpretations

Table 3.1: Summary of the three datasets

Table 4.1: Model artefacts saved for each dataset

Table 5.1: Performance comparison across all nine experiments

Table 5.2: Confusion matrix counts and interpretation

Table 6.1: Feature-importance interpretation

Table 6.2: Predictions on the unlabelled dataset

1. INTRODUCTION

1.1 Background

Churn is not always an event with one clear reason. It can be linked to how sensitive a customer's to price a mismatch between what the customer needs and the plan offered issues with service competing offers or changes in how often the service is used. A predictive model cannot see all the reasons behind a customer's choice. It only learns patterns from the data in the columns. This difference is important because a statistical link in data might not reflect the real reason why a customer left. Telecom data works well for classification tasks because it often includes both data—like call minutes or bill amounts—and categorical data—like the type of plan a customer chose. These different types of information need handling before building a model. They also allow us to compare statistical approaches with tree-based methods that can capture more complex, nonlinear patterns.

Telecom services include voice calls, text messaging, data access and other digital features. Customers may stop using these services due to cost plan fit bad service experience, competition, changing needs or other factors. When a customer stops using a service it is called churn. Understanding churn matters because losing customers affects income and long-term planning. Predictive analytics uses data to guess outcomes for cases we haven't seen yet. In a churn project customer details are the input variables and a churn label is the target.

1.2 Problem Statement

The main issue in this project is to build and test models that classify customer records as either churn or no-churn. The model must be judged not by how many predictions are right but also by the kinds of errors it makes. A false positive happens when a customer who is not going to churn is marked as likely to leave. A false negative occurs when a customer who actually churns is not flagged by the model.

The project also looks at model reuse. A saved classifier should give the results when used later as long as the preprocessing and feature setup stay the same. Re-loading the model and running it again helps confirm that the saved version works for making predictions. The work checks whether standard classification algorithms can separate churn and nonchurn records in telecom data. It compares how well models perform across datasets and tests whether a saved model can be used again to make predictions.

1.3 Motivation

Retention teams usually have limited time and money. A list of customers who look like churners can help them decide who to focus on first.. Model predictions should not be treated as final decisions. Customer background, service history and the cost of retention actions still need to be considered.

Another reason for this work is to compare evaluation methods. A model might seem accurate because most customers don't churn. Still miss many churn cases. Looking at precision, recall, F1-score ROC-AUC and confusion matrix values gives an understanding of how the model behaves. A simple list of churn rates from the past tells us what happened. It doesn't show which individual customers are most likely to leave. Classification models can help rank

customers for follow-up. Still any real system must deal with alerts, missed churn cases, customer privacy, model changes, over time and the cost of retention efforts.

1.4 Objectives

- To define customer churn as a supervised classification problem and establish a clear modelling workflow.
- To inspect and prepare three telecom-related datasets for analysis.
- To train Logistic Regression, Decision Tree, and Random Forest models across the datasets.
- To compare nine model experiments using multiple evaluation metrics rather than accuracy alone.
- To analyse the final model's confusion matrix and discuss false-positive and false-negative outcomes.
- To review important features and explain how they may support customer-retention decisions.

1.5 Scope

The scope includes dataset inspection, preprocessing, feature preparation, exploratory data analysis, model training, quantitative evaluation, basic model interpretation, saved model artefacts, and batch prediction. The project uses three classification approaches that provide different levels of complexity and interpretability. The final Dataset 3 Random Forest is treated as the best-performing model among the experiments recorded for this project.

1.6 Significance

The project demonstrates how predictive modelling can convert historical customer records into a structured risk estimate. A retention team could use a risk score to prioritise review, but the score should be combined with service history, customer preferences, contact rules, and the likely cost of an intervention. For example, a customer predicted to churn may need a service-quality review rather than a discount. The appropriate action depends on the reason for risk and on whether the action is useful to the customer.

The project is also significant as a learning exercise in model selection. Accuracy, precision, recall, F1-score, and ROC-AUC describe different aspects of classification performance. Reading these measures together gives a more balanced view than selecting a model based on a single headline number. The confusion matrix makes the consequences of classification errors explicit and provides a basis for discussing the operational trade-off between missed churners and unnecessary retention contacts.



Figure 1.1: End-to-end workflow of the churn prediction project

2 Literature Review

2.1 Knowledge Discovery in Databases and Predictive Analytics

Knowledge Discovery in Databases (KDD) is a structured process for extracting useful patterns from data. It is commonly described as involving problem understanding, data selection, preprocessing, transformation, data mining, interpretation, and use of the resulting knowledge. These stages are iterative: a model's errors may reveal data-quality issues, while exploratory analysis may suggest a different feature representation. A technically successful model is only one component of KDD; the output must also be interpretable and relevant to the decision being supported.

Predictive analytics focuses on estimating an outcome for new or unseen cases. In supervised learning, a training set contains both explanatory variables and a known target. The algorithm learns a relationship from those examples, and a separate test set estimates how well the learned relationship generalises. Separation between training and test records is essential because evaluating a model on the same examples used to fit it can produce an overly optimistic estimate.

A churn project fits this framework because the target is a customer's churn label and the input variables describe customer characteristics or service usage. The analysis needs to consider data quality, target definition, class balance, and the intended use of predictions. A prediction that is statistically useful but not connected to an appropriate retention process may have little practical value.

2.2 Customer Churn Prediction

Churn prediction has been studied as a classification task in subscription services, telecommunications, banking, and other industries where customers can discontinue a relationship. Typical predictors may include service usage, billing amounts, contract type, support interactions, plan characteristics, and account tenure. The relevance of any particular feature depends on the data source and on the period in which it was measured.

Churn labels are often less common than non-churn labels. This imbalance means that a model can obtain high accuracy by favouring the majority class while failing to identify enough churners. Precision measures how many predicted churners are actual churners, while recall measures how many actual churners are detected. The F1-score balances precision and recall through their harmonic mean. ROC-AUC measures the ability to rank positive cases above negative cases across thresholds, although it does not by itself specify an operational threshold.

2.3 Logistic Regression

Logistic Regression is a widely used baseline for binary classification. It estimates the probability of a class by applying a logistic function to a weighted combination of input features. With suitable encoding, its coefficients can provide a direction and approximate strength of association between a feature and the predicted log-odds, although coefficient interpretation depends on scaling, category encoding, correlation among features, and model specification.

The method is relatively efficient and can be easier to explain than more complex ensembles. It is useful as a benchmark because it provides a clear comparison point. However, a basic Logistic Regression model may not capture complex nonlinear relationships unless interaction

terms or nonlinear transformations are introduced. Its performance also depends on preprocessing choices, such as handling categorical variables and scaling numeric inputs where appropriate.

Algorithm	Main strengths	Potential limitations	Role in this project
Logistic Regression	Efficient baseline; probability output; relatively interpretable coefficients.	May underfit nonlinear patterns; preprocessing and encoding matter.	Baseline classifier across all three datasets.
Decision Tree	Captures nonlinear rules; can be visualised; handles mixed patterns after encoding.	Can overfit; small data changes may alter the tree.	Rule-based comparison model.
Random Forest	Combines many trees; models nonlinear interactions; often robust to individual-tree variance.	Less directly interpretable; larger model artefacts; may need calibration and tuning.	Best recorded model on Dataset 3.

Table 2.1: Comparison of the three machine-learning algorithms

2.4 Decision Tree and Random Forest

A Decision Tree divides the feature space into smaller regions by selecting splits that improve class separation according to a criterion such as Gini impurity or entropy. Its rules can be expressed as a sequence of decisions, which makes the model comparatively intuitive. For example, a path through a tree may combine account or usage characteristics to produce a churn prediction. The tree can nevertheless become too complex and fit noise if its depth, minimum leaf size, or other controls are not selected carefully.

Random Forest is an ensemble method that trains multiple decision trees using randomised samples and feature subsets, then combines their predictions. The averaging process generally reduces the variance of a single tree and can model nonlinear patterns without requiring the analyst to specify every interaction. It can also provide feature-importance summaries. These summaries should be interpreted carefully: importance indicates how a model uses a feature, not necessarily that the feature causes churn.

The recorded experiments show that Random Forest performed particularly well on Dataset 3, achieving the highest F1-score and ROC-AUC among the nine experiments. This does not mean that Random Forest must be best for every telecom dataset. Its relative performance changed across datasets, reinforcing the need to evaluate candidate models on representative data and to consider the intended operating conditions.

2.5 Explainable Machine Learning

Explainable machine learning aims to make model behaviour understandable to people who use, review, or are affected by predictions. Explanation can be global, describing patterns used across the dataset, or local, explaining one prediction. Common approaches include model coefficients, decision rules, permutation importance, partial dependence, and local explanation

methods such as SHAP. Each method answers a different question and can have limitations when features are correlated or when the model is applied outside its training distribution.

In a churn workflow, feature-importance results can help analysts ask better questions: Are service complaints associated with elevated risk? Do usage patterns differ between groups? Are plan characteristics acting as proxies for another factor? The explanation should be checked against domain knowledge and data quality. A feature ranking should not be used as proof of causation, and customer-facing actions should not be based solely on an opaque score.

2.6 Summary and Relevance

The literature and standard predictive-modelling principles support a workflow that begins with business understanding, maintains a clear separation between training and test data, compares multiple algorithms, and reports class-sensitive metrics. Logistic Regression provides a baseline, Decision Tree offers rule-based structure, and Random Forest provides an ensemble capable of learning more complex relationships. Explainability provides a bridge between model output and responsible use.

These concepts directly inform the present project. The experiments are compared across three datasets, the final model is assessed with a confusion matrix and several metrics, and feature importance is discussed as an aid to interpretation. The project also recognises that the strongest experimental score does not automatically establish production readiness.

Metric	Meaning	Why it matters for churn
Accuracy	Fraction of all predictions that are correct.	Useful overall summary, but can hide poor detection of a minority class.
Precision	Fraction of predicted churners who truly churned.	Higher precision can reduce unnecessary retention outreach.
Recall	Fraction of actual churners correctly detected.	Higher recall can help identify more at-risk customers.
F1-score	Harmonic mean of precision and recall.	Useful when both missed churners and false alarms matter.
ROC-AUC	Ranking discrimination across thresholds.	Summarises how well positive cases tend to rank above negative cases.

Table 2.2: Evaluation metrics and their interpretations

3 Workflow / Predictive Modelling Process

3.1 Overview of the Workflow

The project follows a staged predictive-modelling process. The stages are presented in a logical order, but the work is iterative: findings from evaluation may lead to additional preprocessing or feature checks. Each stage has a specific purpose and produces an output that supports the next stage. Keeping this structure helps make the analysis reproducible and makes it easier to explain where a modelling decision was made.

The workflow begins by defining churn as the target and identifying how a prediction might be used. The datasets are then inspected for row counts, column types, target availability, missing values, duplicates, and unusual values. Preprocessing prepares the data for modelling, feature preparation converts fields into model-compatible inputs, and exploratory analysis examines distributions and possible relationships. The three algorithms are trained, evaluated, compared, and used to generate predictions for unlabelled records.



Figure 1.1: End-to-end workflow of the churn prediction project

3.2 Stage 1: Problem Identification and Understanding

The first stage identifies the decision that the model is intended to support. Here, the outcome is whether a telecom customer is labelled as having churned. The task is therefore a binary classification problem. A useful prediction should help a business decide which accounts deserve additional review, rather than automatically assuming that every flagged customer will leave.

Problem definition also requires a clear target convention. The analyst must know which label represents churn, whether the label is recorded as text or a numeric value, and which date or period it covers. If features are recorded after a customer has already churned, they may leak information about the target and make the evaluation unrealistic. The project therefore treats target definition and feature timing as important checks before model training.

The business objective influences metric choice. If a retention team can contact only a limited number of customers, precision among the highest-risk cases may be important. If the cost of missing churners is especially high, recall may receive greater emphasis. The project reports several metrics so that different priorities can be discussed rather than silently assuming a single business cost structure.

3.3 Stage 2: Data Selection and Understanding

Three datasets were examined to compare model behaviour across different data sources. The IBM Telco Customer Churn dataset provides a familiar benchmark with 7,043 customer records. The second telecom dataset contains approximately 51,047 records. The third dataset was used for final model selection and included a labelled training portion and a held-out test

portion; the recorded test set contained 850 records, while a separate unlabelled file contained 750 records for batch prediction.

Dataset size alone does not determine usefulness. A larger dataset may contain more examples but can differ in target definition, feature availability, collection process, and noise. For that reason, scores from different datasets should be interpreted as separate experiments rather than as a perfectly controlled comparison. The same algorithm may perform differently because the datasets represent different customer populations or encode churn differently.

Dataset	Recorded size / role	Use in the project	Important caution
IBM Telco Customer Churn	7,043 records	Benchmark modelling dataset.	Check target encoding, missing values, and categorical fields.
Telecom Data	Approximately 51,047 records	Larger dataset for comparative experiments.	Dataset schema and label definition may differ from IBM Telco.
Dataset 3	Labelled train/test split; 850 test records	Final model comparison and evaluation.	Confirm feature/label consistency and preserve a held-out test set.
Telecom - Prediction Data	750 unlabelled records	Batch inference after model selection.	Predictions are estimates; true outcomes are not available for validation.

Table 3.1: Summary of the three datasets

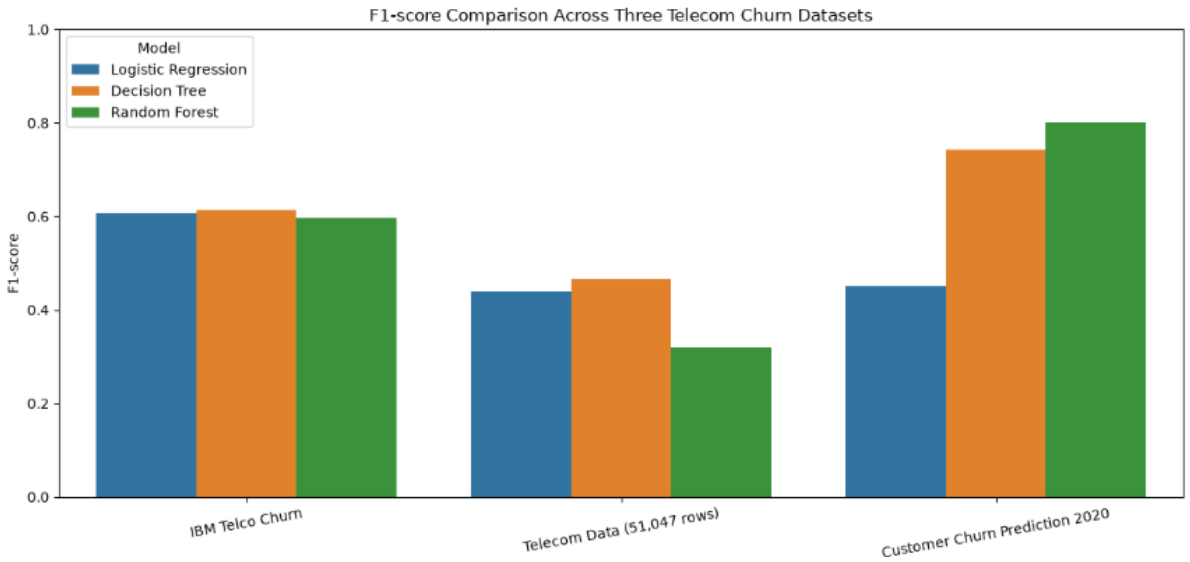


Figure 3.1: Record counts for the three analysed datasets

The record-count figure is intended to communicate the relative scale of the data sources. It should not be interpreted as evidence that the datasets are directly comparable in quality or

content. In the final implementation, dataset-specific preprocessing and target mapping must be preserved alongside the trained estimator.

3.4 Stage 3: Data Preprocessing

Preprocessing converts raw files into a consistent representation suitable for analysis. The first checks typically include reading the file successfully, reviewing dimensions and column names, inspecting data types, identifying missing values, checking duplicated records, and summarising numerical fields. These checks help identify malformed columns, unexpected text values, inconsistent categories, and records that could distort the analysis.

Categorical variables usually need to be encoded before they can be used by standard scikit-learn estimators. One-hot encoding is a common approach for nominal categories because it creates indicator columns without imposing an artificial order. Binary categories can be mapped to consistent numeric values when the meaning is unambiguous. Numerical fields may require conversion from text, and values such as blank strings or special tokens should be treated consistently rather than accidentally interpreted as valid numbers.

Preprocessing decisions must be learned from the training data wherever possible. For example, an imputer or scaler should be fitted on the training partition and then applied to validation or test data. This prevents information from the test set from influencing the training process. If preprocessing is implemented in a scikit-learn pipeline, the sequence can be stored with the estimator and reused during inference, reducing the risk of training-serving mismatch.

Duplicate removal requires judgement. Identical rows can represent accidental duplication, but repeated customer records can also be legitimate observations at different times. The decision should depend on the dataset's unit of analysis and any available customer identifier. Similarly, outliers should not be removed merely because they are extreme; unusually high usage or billing may be genuine and relevant to churn.

3.5 Stage 4: Feature Preparation

Feature preparation selects the input columns and transforms them into a form that the model can learn from. The target label must be excluded from the input matrix, as must identifiers that merely name a row unless they have a defensible predictive meaning. Features should be checked for leakage, such as a status field that is created only after churn occurs. A random identifier can create spurious patterns and typically should not be treated as a meaningful behavioural signal.

Numeric features may include charges, minutes, service-call counts, tenure, or usage totals, depending on the dataset. Categorical features may include plan or service indicators. A feature set should be built from fields available at the time the prediction would be made. If a field is unavailable in real operations, its presence during training can make the model look stronger than it would be in practice.

The project compares model families with different feature-response assumptions. Logistic Regression benefits from consistent encoding and may benefit from scaling. Tree-based methods do not generally require feature scaling, although they still require consistent handling

of missing and categorical data. A reproducible pipeline should apply the correct transformations to both training data and new prediction records.

3.6 Stage 5: Exploratory Data Analysis

Exploratory Data Analysis (EDA) helps the analyst understand the dataset before relying on model scores. It includes examining the distribution of the target label, the range and spread of numeric variables, category frequencies, missingness patterns, and potential relationships between features and churn. Visualisations can reveal imbalance, unusual values, or groups that deserve further investigation.

Class distribution is particularly important. When non-churn customers are more common than churn customers, a classifier may achieve a strong accuracy score by mostly predicting no churn. The labelled Dataset 3 training distribution was reviewed to understand this issue. The chart below provides an illustrative visual summary of the class distribution used in the project documentation; the underlying counts should be checked against the final notebook before a formal submission if exact train-split counts are required.

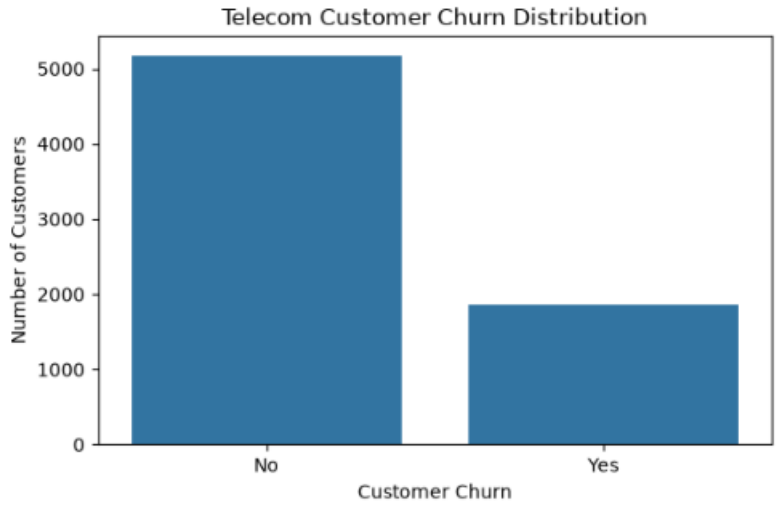


Figure 3.2: Dataset 3 class distribution illustration; confirm exact training counts in the notebook

EDA should also examine numeric features such as usage, billing amounts, and service-contact counts. A relationship between high service-contact frequency and churn, for example, could indicate that unresolved issues are associated with elevated risk. However, a visual association does not prove that the feature causes churn. The pattern may reflect a third factor, an operational policy, or how the data were collected.

Comparisons between datasets should be cautious. Similar field names do not guarantee identical units or definitions. A charge variable may be measured per month in one dataset and over a different interval in another. Before interpreting distributions, the analyst should confirm the field meaning, units, date coverage, and target definition. This prevents a visually convincing but invalid comparison.

3.7 Stage 6: Predictive Modelling

Three classifiers were included: Logistic Regression, Decision Tree, and Random Forest. Each model was fitted to prepared training data and evaluated using a consistent set of metrics. This

produces nine experiments in total—three algorithms across three datasets. Keeping the algorithm families consistent provides a useful comparative structure, while the different dataset properties remain an important limitation when interpreting cross-dataset differences.

Logistic Regression serves as a baseline because it estimates class probabilities with a relatively simple functional form. Decision Tree can learn nonlinear decision rules and interactions, but an unconstrained tree may overfit. Random Forest combines multiple trees and can reduce the instability of a single tree. In practice, model performance depends on hyperparameters, class balance, feature quality, and the way data are split.

The trained estimators were saved as joblib artefacts so that they could be reloaded without repeating model fitting. Saving a model is only part of reproducibility: the associated feature order, preprocessing steps, package versions, and target mapping should also be documented. A saved estimator should not be used with arbitrary columns or a different schema.

3.8 Stage 7: Model Evaluation

The evaluation stage compares predictions with known labels in a held-out test set. Accuracy is the share of all cases classified correctly. Precision focuses on predicted churners and answers how many of those flags are correct. Recall focuses on actual churners and answers how many the model identifies. F1-score combines precision and recall, while ROC-AUC evaluates the model's ranking ability across thresholds.

No single metric is sufficient for every retention strategy. A high-recall model may find more churners but create more false alarms. A high-precision model may produce a smaller, more reliable list while missing additional at-risk customers. The best operating point depends on the available retention budget, the cost of outreach, and the value of retaining a customer. Threshold selection should therefore be a business decision informed by validation data.

A held-out test set should ideally be used once for final assessment, after the model and threshold have been selected using training or validation data. If many experiments are repeatedly tuned against the test set, the test score can gradually become optimistic. For production use, temporal validation may be more realistic than a random split because the model will be applied to future customers rather than to randomly sampled records from the same period.

3.9 Stage 8: Interpretation and Prediction

After evaluation, the analyst reviews the confusion matrix and feature-importance output to understand the model's behaviour. The confusion matrix separates correct no-churn predictions, correct churn predictions, false alarms, and missed churners. Feature importance gives a high-level view of which input variables the model uses most strongly, but it should be treated as descriptive rather than causal.

The selected Dataset 3 Random Forest was applied to 750 unlabelled records. The prediction file contains an identifier and a predicted churn label, with 649 records classified as no churn and 101 classified as churn.

For an operational workflow, the prediction output should ideally include a timestamp, model version, probability or risk score, and a clear definition of the positive class. A review process should also be established so that customers are not automatically penalised because of a prediction. The prediction should trigger a proportionate, customer-benefiting review rather than an unquestioned action.

4 Results and Discussion

4.1 Model Performance

The project recorded nine experiments across three datasets and three algorithms. The results show that performance depends on both the dataset and the algorithm. On the IBM Telco dataset, the Decision Tree achieved an F1-score of approximately 0.615, slightly higher than Logistic Regression at approximately 0.608 and Random Forest at approximately 0.598. On the larger Telecom Data dataset, Random Forest achieved the highest accuracy among its three experiments but had a lower recall and F1-score, showing why accuracy alone can be misleading.

Dataset 3 produced the strongest final result. Its Random Forest reached an accuracy of 0.9482, precision of 0.8800, recall of 0.7333, F1-score of 0.8000, and ROC-AUC of 0.8887. The Decision Tree reached an F1-score of 0.7436, while Logistic Regression reached 0.4514. The Random Forest therefore offered the best balance of precision and recall among the nine recorded experiments and was selected for the final demonstration.

The results should not be interpreted as a controlled ranking of dataset quality. The datasets differ in their size and potentially in their labels, feature definitions, and population. The experiment table is most useful as a record of how each candidate performed within its own dataset and as a reminder that model choice should be validated against the data expected in the intended deployment.

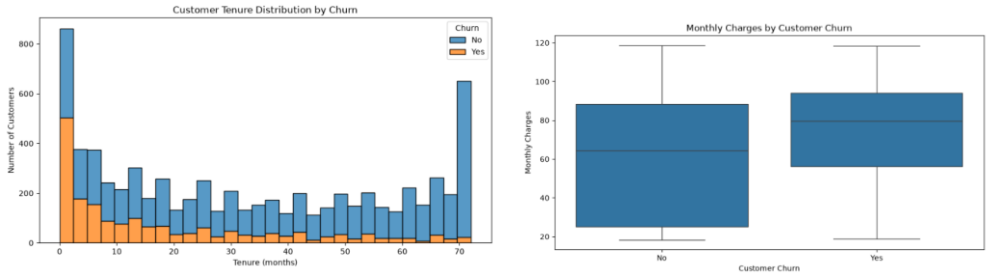


Figure 5.1: F1-score comparison across nine experiments

Dataset	Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
IBM Telco	Logistic Regression	0.8038	0.6485	0.5722	0.6080	0.8359
IBM Telco	Decision Tree	0.7335	0.4992	0.7995	0.6146	0.8182
IBM Telco	Random Forest	0.7676	0.5535	0.6497	0.5978	0.8146
Telecom Data	Logistic Regression	0.5793	0.3566	0.5717	0.4392	0.6117
Telecom Data	Decision Tree	0.5920	0.3736	0.6145	0.4647	0.6319
Telecom Data	Random Forest	0.7071	0.4831	0.2379	0.3188	0.6586
Dataset 3	Logistic Regression	0.7541	0.3295	0.7167	0.4514	0.7926
Dataset 3	Decision Tree	0.9294	0.7632	0.7250	0.7436	0.8151
Dataset 3	Random Forest	0.9482	0.8800	0.7333	0.8000	0.8887

Table 5.1: Performance comparison across all nine experiments

4.2 Final Model Confusion Matrix

The Dataset 3 Random Forest confusion matrix provides a more detailed account of the final test performance. Of 730 actual no-churn cases, 718 were correctly predicted as no churn and

12 were incorrectly predicted as churn. Of 120 actual churn cases, 88 were correctly detected and 32 were missed. The matrix therefore distinguishes correct classifications from the two types of error that matter to a retention team.

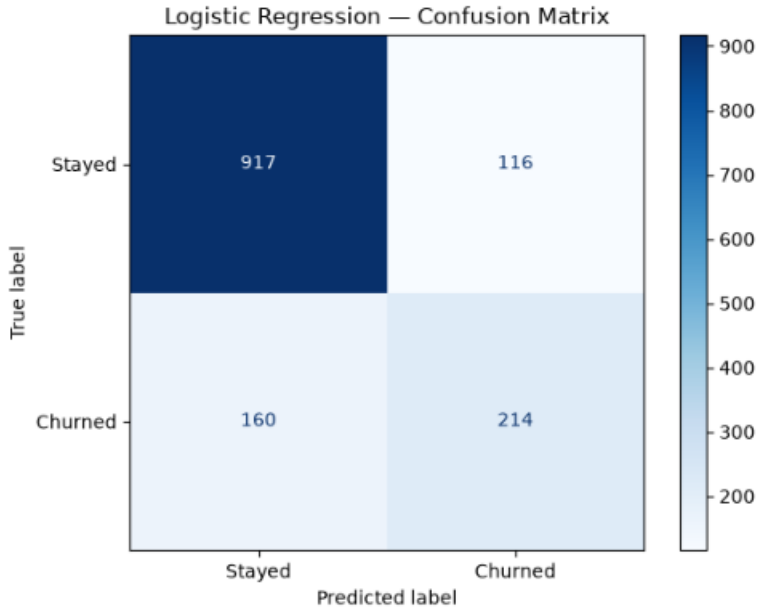


Figure 5.3: Confusion matrix of the final Random Forest

Outcome	Count	Interpretation
True Negative (TN)	718	Actual no-churn customers correctly predicted as no churn.
False Positive (FP)	12	Customers predicted to churn who were labelled no churn.
False Negative (FN)	32	Actual churners incorrectly predicted as no churn.
True Positive (TP)	88	Actual churners correctly identified by the model.

Table 5.2: Confusion matrix counts and interpretation

The 12 false positives represent a relatively small number of unnecessary flags in this test set. The 32 false negatives show that the model still missed some customers who churned. In a real retention programme, the cost of these errors would depend on the value of the customer, the cost of the offer, the likelihood that outreach changes the outcome, and the effect of unnecessary contact on customer experience.

The metrics are consistent with the confusion matrix. Precision is 88%, meaning that most customers flagged as churners in this test set were labelled as churners. Recall is 73.33%, meaning that the model identified roughly three quarters of the actual churn cases. The F1-

score of 80% reflects the balance between those two measures. The result is promising for a project baseline, but additional validation is needed before operational use.

4.3 Feature Importance

Feature importance is used to explore which input variables contributed most to the Random Forest's predictions. The project's recorded interpretation highlighted total day charges, customer service calls, total day minutes, international-plan indicators, and evening usage or charges. These variables can be viewed as signals that may help distinguish customer groups in the dataset.

Higher day charges may be associated with a customer's usage level, plan fit, or perceived value. Repeated customer-service calls may indicate friction or unresolved issues. Day minutes and evening usage may capture consumption patterns that interact with plan design. An international-plan indicator can reflect a service choice that changes the cost structure or the customer's expectations. These are plausible interpretations, not proof that the feature causes churn.

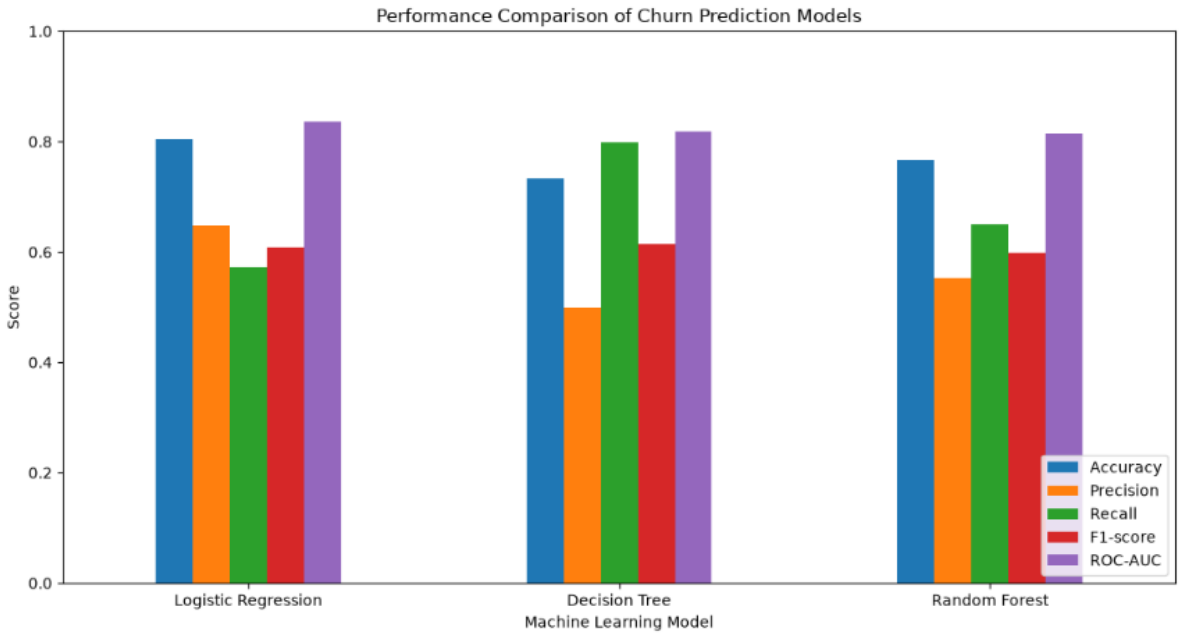


Figure 6.1: Qualitative overview of leading features (relative prominence is illustrative, not measured importance)

The figure is explicitly qualitative because exact feature-importance percentages were not provided in the project record. It should not be used as a numerical ranking or as a substitute for the saved feature-importance output. In a final empirical report, the exact top features and values should be copied from the model's exported feature-importance file and checked against the model's feature names.

Feature importance can also be biased by the type and distribution of features. Tree-based impurity importance may favour variables with many possible split points, and correlated variables can divide importance between them. Permutation importance on a held-out set can provide a complementary view, while SHAP-style explanations can help analyse individual

predictions. Explanations should be stable, validated, and phrased in a way that avoids implying causality.

4.4 Prediction Demonstration

After the final model was selected, it was used to predict churn labels for a separate file containing 750 unlabelled records. The resulting output file, `dataset3_predictions.csv`, contains two columns: `id` and `churn`. The recorded prediction counts were 649 no-churn cases and 101 churn cases. This output demonstrates the inference stage of the workflow: a trained estimator is applied to new records that do not contain known outcomes.

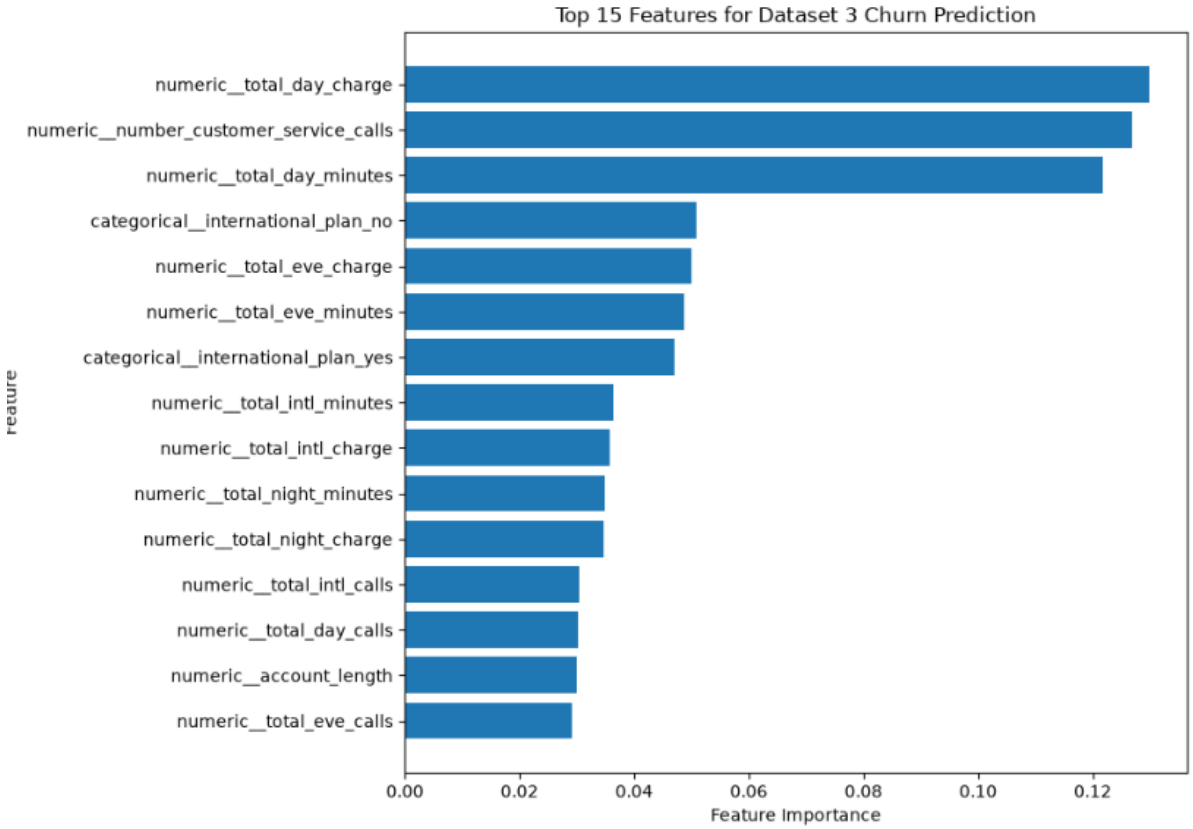


Figure 6.2: Predictions for 750 unlabelled records

4.4 Prediction Demonstration (continued)

The predicted class counts should not be described as a measured churn rate for the full customer population unless the unlabelled file is known to be representative. The model may be applied to a sample selected for a particular purpose, and its output depends on the feature distribution, model calibration, and chosen decision threshold. Since no ground-truth labels are available for these 750 records, the quality of these predictions cannot be directly measured from this file alone.

For practical review, a team could sort records by predicted churn probability, inspect the highest-risk group, and compare the proposed intervention with the likely reason for risk. Probability estimates should be checked for calibration before they are interpreted as literal

probabilities. A prediction such as 0.80 does not automatically mean that exactly 80 out of 100 comparable customers will churn unless the model is well calibrated on relevant data.

4.5 Practical Implications

The strongest practical use of the model is prioritisation. A churn-risk score can help a team focus attention on customers who may benefit from a service review, a plan adjustment, or proactive support. It can also support aggregate analysis, such as identifying whether particular service combinations or usage patterns are associated with elevated churn. Any action should be selected based on customer needs and operational evidence, not solely on the predicted label.

A responsible retention process should define who can access scores, how long scores are retained, which interventions are permitted, and how outcomes are measured. The organisation should compare contacted and comparable non-contacted customers where feasible to estimate whether an intervention actually changes churn. Without such evaluation, a model may identify customers who were already likely to leave but fail to show that a particular action prevents churn.

The model should also be monitored after deployment. Customer behaviour, pricing, network performance, plan offerings, and competitor activity can change over time. These changes can cause data drift or concept drift, reducing predictive performance. Periodic evaluation should track precision, recall, calibration, false-positive rates, and false-negative rates, with retraining considered only after data quality and changes in the business process have been reviewed.

4.6 Limitations

The project has several limitations. First, the three datasets may use different feature definitions and label-generation processes, so the scores should not be compared as if the data were identical. Second, the recorded evaluation uses a held-out test set for Dataset 3, but a single split may not capture variation across time or customer segments. Third, the model's performance depends on preprocessing and feature availability, and a production environment may differ from the experimental environment.

Fourth, feature importance is descriptive and does not establish causality. Fifth, the 750 unlabelled predictions cannot be verified without subsequent outcome labels. Sixth, the report does not establish that a retention action based on the model will reduce churn; that requires an outcome evaluation or controlled intervention. Finally, a high accuracy score can be sensitive to class balance and the decision threshold, so future work should include threshold analysis, calibration, and a more explicit cost-sensitive evaluation.

5 Conclusion

5.1 Summary of the Work

This project developed a structured workflow for telecom customer churn prediction and compared Logistic Regression, Decision Tree, and Random Forest across three datasets. The process included problem definition, dataset inspection, preprocessing, feature preparation, exploratory analysis, model training, evaluation, interpretation, and batch prediction. Nine model experiments were documented to provide a consistent view of how the algorithms performed on the selected datasets.

The final Dataset 3 Random Forest achieved the strongest recorded overall balance of metrics, with 94.82% accuracy, 88.00% precision, 73.33% recall, an F1-score of 80.00%, and ROC-AUC of 0.8887. Its confusion matrix showed 718 true negatives, 12 false positives, 32 false negatives, and 88 true positives. The model was then applied to 750 unlabelled records and classified 101 as churn and 649 as no churn.

5.2 Key Findings

- Model performance varied substantially across datasets, so dataset context matters as much as algorithm choice.
- Accuracy alone did not provide a complete picture; precision, recall, F1-score, ROC-AUC, and the confusion matrix were needed for a more balanced assessment.
- Dataset 3 Random Forest was the best recorded experiment according to the project's combined evaluation results.
- The model identified a majority of the churn cases in the held-out test set, but 32 actual churners were still missed.
- Feature signals such as charges, usage, service calls, and plan indicators can support investigation, but they should not be interpreted as causes without further evidence.
- Predictions for unlabelled records are useful for demonstrating inference, but cannot be scored until actual outcomes become available.

5.3 Future Work

Future work should first verify the final data pipeline end to end, including column names, target mapping, missing-value treatment, and consistent preprocessing between training and prediction. A stratified split can help preserve class proportions, while temporal validation can provide a stronger estimate of performance on future customer cohorts. Repeated cross-validation can estimate the variability of the results rather than relying on one split.

The model could be improved through hyperparameter tuning, probability calibration, threshold selection based on business costs, and comparison with additional algorithms such as gradient boosting. Explainability could be strengthened by reporting exact feature-importance values from the saved artefact and adding local explanations for individual cases. Future evaluation should also examine performance across customer groups to identify systematic differences or unfair error patterns.

Finally, the project could be extended into a monitored application that accepts a validated customer record, returns a risk score, and records model version and prediction time. Any real-world pilot should measure actual churn outcomes and test whether the chosen intervention changes them. A model should be retained only if it provides reliable, measurable value while respecting privacy, transparency, and customer interests.

6 References

1. Fayyad, U., Piatetsky-Shapiro, G., & Smyth, P. (1996). From data mining to knowledge discovery in databases. AI Magazine, 17(3), 37–54.

2. Breiman, L. (2001). Random forests. Machine Learning, 45, 5–32.

3. Quinlan, J. R. (1986). Induction of decision trees. Machine Learning, 1, 81–106.

4. Cox, D. R. (1958). The regression analysis of binary sequences. Journal of the Royal Statistical Society: Series B, 20(2), 215–242.

5. Fawcett, T. (2006). An introduction to ROC analysis. Pattern Recognition Letters, 27(8), 861–874.

6. Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

7. IBM. Telco Customer Churn dataset. IBM Sample Data Sets. Dataset source and access details should be verified against the version used in the project.

8. Scikit-learn developers. Scikit-learn user guide: Supervised learning, model evaluation, preprocessing, and ensemble methods. https://scikit-learn.org/stable/user_guide.html

9. Molnar, C. (2022). Interpretable Machine Learning (2nd ed.). https://christophm.github.io/interpretable-ml-book/

10. Joblib developers. Joblib documentation: Persistence utilities for Python objects. https://joblib.readthedocs.io/

11. Project artefacts: telecom churn modelling notebook, saved model files, model comparison CSV, classification report, confusion matrix, and feature-importance output maintained with the project files.

APPENDIX A: REPRODUCIBILITY AND MODEL ARTEFACTS

Reproducibility means that another analyst can understand the data used, repeat the preprocessing, reload the estimator, and obtain predictions using the same feature schema. A saved model file alone is not enough if the preprocessing logic or expected feature order is

missing. The project therefore maintains model artefacts alongside a notebook and evaluation reports.

Dataset	Saved model artefacts	Purpose
Dataset 1	logistic_regression.joblib; decision_tree.joblib; random_forest.joblib	Persist the three benchmark classifiers.
Dataset 2	dataset2_logistic_regression.joblib; dataset2_decision_tree.joblib; dataset2_random_forest.joblib	Persist the three models trained on the larger dataset.
Dataset 3	dataset3_logistic_regression.joblib; dataset3_decision_tree.joblib; dataset3_random_forest.joblib	Persist the three models used for the final comparison.

Table 4.1: Model artefacts saved for each dataset

The accompanying project outputs include the main notebook, a nine-experiment comparison CSV, final Dataset 3 metrics, confusion-matrix counts, a classification report, a feature-importance export, and the 750-record prediction file. These outputs provide a traceable record of model evaluation and inference.

For a repeatable run, the analyst should document Python and library versions, the source file names, the target column, feature columns, split strategy, random seed, and any preprocessing choices. Model files should be loaded only from a trusted source because serialised Python objects can execute code during deserialisation. Large artefacts should be stored in a location suitable for their size, and the repository should avoid committing generated files that exceed hosting limits.

A final verification checklist should include: confirming that the notebook runs from a clean environment; checking that all required data files are present; confirming the target label mapping; verifying that training and inference use the same feature order; reproducing the final

metrics; checking that no test labels entered training; and confirming that unlabelled predictions contain exactly the expected number of records.

APPENDIX B: RESPONSIBLE USE CHECKLIST

- A churn model can influence how customers are treated, so its use should be proportionate and transparent. The following checklist summarises safeguards that should accompany any attempt to use the project beyond an academic demonstration.
- Purpose limitation: use predictions only for a clearly defined retention or service-improvement purpose.
- Data minimisation: include only fields that are relevant and appropriately authorised for the intended analysis.
- Quality checks: review missingness, label quality, data drift, and schema changes before each scoring run.
- Human review: use the model to prioritise review rather than automatically making consequential customer decisions.
- Error monitoring: track false positives and false negatives, including differences across meaningful customer groups.
- Outcome evaluation: measure whether interventions improve retention rather than assuming that a prediction itself creates value.
- Security: restrict access to customer records, model files, and prediction outputs; follow organisational retention rules.
- Communication: explain that scores are estimates and avoid presenting a model output as certainty.
- Revalidation: retrain or recalibrate only after reviewing current data, performance, and the reasons for any degradation.

Overall, the project provides a practical baseline for exploring churn prediction with explainable machine-learning methods. Its most important contribution is not only the final score but also the documented comparison, the interpretation of classification errors, and the recognition that prediction quality, business value, and responsible use must be considered together.